

Airframe Technology Master

Lecture 13.

Modelling Guidelines for Interfaces: MPCs & RBEs. Fittings. Connectors. Movable Surfaces

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Objective

The primary objective is:

- to describe the modelling guidelines for FEM interfaces
- to identify the main NASTRAN element options for the interfaces

The main questions that the student should answer to the end of this class are:

- Structural interfaces
 - What is an interface?
 - Why is important to capture the interface loads?
 - Interface typical GFEM modelling
- What are the element options to model an interface in a FEM and to obtain the interface loads results.
- What are the differences between CELAS & CBUSH elements (typical interface elements)
- What are the main functions and differences of explicit MPC, RBE2 & RB3
- Example of movable surface FEM interface

Overview

- Structural interfaces
- CELAS / CBUSH: NASTRAN interface elements
 - Coincident Nodes
 - Coordinate systems
 - Interface Stiffness
- MPC's and RBEs: Degrees of Freedom Relations
- Concept of degree of freedom
 - Nastran singularities in execution process
- Explicit MPC: Definition & typical usage
- Main RB elements: Definition & typical usage
 - RBE2
 - RBE3
- Application to Movable Surfaces modelling

Interfaces

— Interface Definition:

- The interfaces are the locations where 2 structural components are joined each other.
- Typically the interfaces are defined as junctions in discrete locations.
- A typical requirement for the interfaces is that the joined structural components must be replaceable or interchangeable. In general terms that means that the assembly/disassembly operations must be quick and simple.
- The most typical structural configuration for an interface is a lug / clevis joint

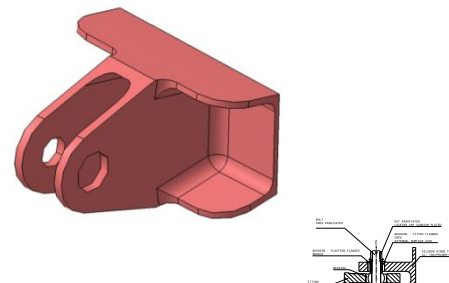
— Interface Loads:

- The interface loads are the forces/moments applied by one structural component on the other through the interface. The interfaces are the load path between the two components.
- Typically different work teams or companies are in charge of each component. For that reason the **interface loads are very important for a correct evolution of the work developed by each team or company**

— Typical aircraft component interfaces:

- Wing/Fuselage & Flap/Wing & Aileron/Wing
- HTP/Fuselage & Elevator/HTP
- VTP/Fuselage & Rudder/VTP
- Pylon/Wing & Nacelle/Pylon

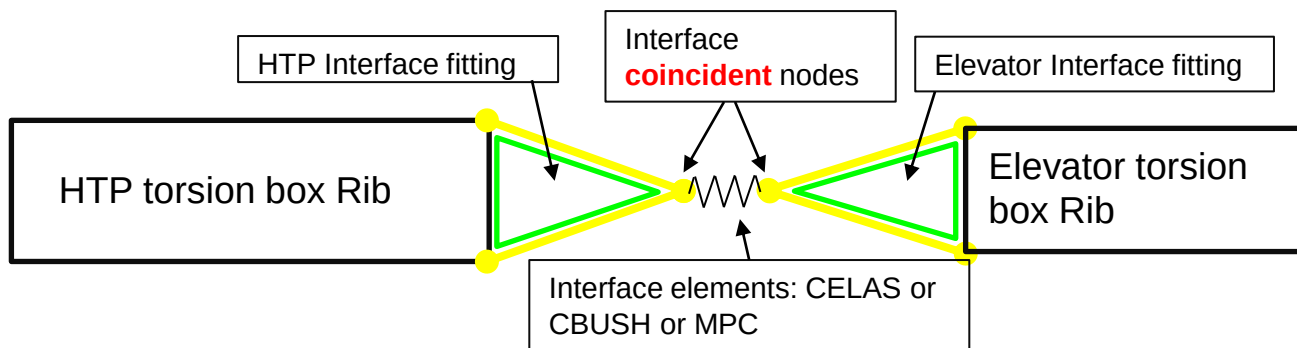
- Most of these interfaces are a lug / clevis joint



Interfaces (cont'd)

— Interfaces typical GFEM modelling:

- Main objective of a GFEM is to model correctly the load paths and hence the interface loads.
- The interface modelling must ensure a correct representation of the interface stiffness.
- A typical interface model is:
 - Interface nodes: 2 coincident nodes located at the agreed interface coordinates (1 node per each structural component)
 - Interface elements: The 2 interfaces nodes connected by means of CELAS or CBUSH or explicit MPC (NASTRAN entities)
 - Interface loads are provided by these interface elements
 - CELAS & CBUSH provide the option of defining the interface local stiffness.
 - MPC establishes relations among the degree of freedom of both nodes.
 - Interfaces fittings modelled either with:
 - RBE2 elements: they are rigid elements with infinite stiffness, that are used if the fitting stiffness is much higher than the adjacent structure. It is the simplest model.
 - CTRIA/CQUAD/CROD: they enable to reproduce more accurately the fitting stiffness.



CELAS1 & CELAS2 : NASTRAN interface Elements

CELASx are NASTRAN elements that connect the DOF of 2 nodes, as a spring. DOF relationship is established based on the analysis coordinate systems of the two nodes

CELAS1 Scalar Spring Connection

Defines a scalar spring element.

Format:

1	2	3	4	5	6	7	8	9	10
CELAS1	EID	PID	G1	C1	G2	C2			

Field	Contents
EID	Unique element identification number. ($0 < \text{Integer} < 100,000,000$)
PID	Property identification number of a PELAS entry. (Integer > 0 ; Default = EID)
G1, G2	Geometric grid point identification number. (Integer ≥ 0)
C1, C2	Component number. ($0 \leq \text{Integer} \leq 6$; blank or zero if scalar point.)

PELAS Scalar Elastic Property

Specifies the stiffness, damping coefficient, and stress coefficient of a scalar elastic (spring) element (CELAS1 or CELAS3 entry).

Format:

1	2	3	4	5	6	7	8	9	10
PELAS	PID1	K1	GE1	S1	PID2	K2	GE2	S2	

Field	Contents
PIDi	Property identification number. (Integer > 0)
Ki	Elastic property value. (Real)
GEi	Damping coefficient, g_e . See Remarks 5. and 6. (Real)
Si	Stress coefficient. (Real)

CELAS2 Scalar Spring Property and Connection

Defines a scalar spring element without reference to a property entry.

Format:

1	2	3	4	5	6	7	8	9	10
CELAS2	EID	K	G1	C1	G2	C2	GE	S	

Field	Contents
EID	Unique element identification number. ($0 < \text{Integer} < 100,000,000$)
K	Stiffness of the scalar spring. (Real)
G1, G2	Geometric grid point or scalar identification number. (Integer ≥ 0)
C1, C2	Component number. ($0 \leq \text{Integer} \leq 6$; 0 blank or zero if scalar point.)
GE	Damping coefficient. See Remarks 6. and 8. (Real)
S	Stress coefficient. (Real)

CELAS NASTRAN interface Elements

CELAS Force Output:

$$F(\text{CELAS}) = K_{\text{dof}_i} * [U(\text{node1})_{\text{dof}_i} - U(\text{node2})_{\text{dof}_i}]$$

Where

- K_{dof_i} = Stiffness defined in PELAS/CELAS card for the degree of freedom i.
- $U(\text{node1})_{\text{dof}_i}$ = displacement of the node 1 in the direction of the dof 1, according to the analysis coordinate system of the node 1
- $U(\text{node2})_{\text{dof}_i}$ = displacement of the node 2 in the direction of the dof 1, according to the analysis coordinate system of the node 2

REMINDER:

- CELAS force sign depends on how the nodes have been ordered in defining the element.
- CELAS force direction is defined through the node analysis coordinate system

Recommendations for CELAS creation:

- CELAS null length
- Identical analysis coordinate system for both nodes
- Directions of the analysis system for the nodes must have physical sense. (f.i: bolt tension and shear directions)
- Identical criteria choosing node 1 (structure A) and node 2 (structure B) in all the interface CELAS elements (easier force post-processing)

CBUSH: NASTRAN interface Elements

CBUSH are NASTRAN elements that connect the DOF of 2 nodes, as a spring. DOF relationship can be established based on any coordinate system

CBUSH Generalized Spring-and-Damper Connection

Defines a generalized spring-and-damper structural element that may be nonlinear or frequency dependent.

Format:

1	2	3	4	5	6	7	8	9	10
CBUSH	EID	PID	GA	GB	GO/X1	X2	X3	CID	
	S	OCID	S1	S2	S3				

Field	Contents
EID	Element identification number. (0 < Integer < 100,000,000)
PID	Property identification number of a PBUSH entry. (Integer > 0; Default = EID)
GA, GB	Grid point identification number of connection points. See Remark 6. (Integer > 0)
Xi	Components of orientation vector $\vec{v}$, from GA, in the displacement coordinate system at GA. (Real)
GO	Alternate method to supply vector $\vec{v}$ using grid point GO. Direction of $\vec{v}$ is from GA to GO. $\vec{v}$ is then transferred to End A. See Remark 3. (Integer > 0)
CID	Element coordinate system identification. A 0 means the basic coordinate system. If CID is blank, then the element coordinate system is determined from GO or Xi. See Figure 8-17 and Remark 3. (Integer ≥ 0 or blank)
S	Location of spring damper. See Figure 8-17. (0.0 ≤ Real ≤ 1.0; Default = 0.5)
OCID	Coordinate system identification of spring-damper offset. See Remark 9. (Integer ≥ -1; Default = -1, which means the offset point lies on the line between GA and GB according to Figure 8-17)
S1, S2, S3	Components of spring-damper offset in the OCID coordinate system if OCID ≥ 0. See Figure 8-18 and Remark 9. (Real)

PBUSH Generalized Spring-and-Damper Property

Defines the nominal property values for a generalized spring-and-damper structural element.

Format:

1	2	3	4	5	6	7	8	9	10
PBUSH	PID	"K"	K1	K2	K3	K4	K5	K6	
		"B"	B1	B2	B3	B4	B5	B6	
		"GE"	GE1	GE2	GE3	GE4	GE5	GE6	
		"RCV"	SA	ST	EA	ET			

Field	Contents
PID	Property identification number. (Integer > 0)
"K"	Flag indicating that the next 1 to 6 fields are stiffness values in the element coordinate system. (Character)
Ki	Nominal stiffness values in directions 1 through 6. See Remarks 2. and 3. (Real; Default = 0.0)
"B"	Flag indicating that the next 1 to 6 fields are force-per-velocity damping. (Character)
Bi	Nominal damping coefficients in direction 1 through 6 in units of force per unit velocity. See Remarks 2., 3., and 9. (Real; Default = 0.0)
"GE"	Flag indicating that the next fields, 1 through 6 are structural damping constants. See Remark 7. (Character)
GEi	Nominal structural damping constant in directions 1 through 6. See Remarks 2. and 3. (Real; Default = 0.0)
"RCV"	Flag indicating that the next 1 to 4 fields are stress or strain coefficients. (Character)
SA	Stress recovery coefficient in the translational component numbers 1 through 3. (Real; Default = 1.0)
ST	Stress recovery coefficient in the rotational component numbers 4 through 6. (Real; Default = 1.0)
EA	Strain recovery coefficient in the translational component numbers 1 through 3. (Real; Default = 1.0)
ET	Strain recovery coefficient in the rotational component numbers 4 through 6. (Real; Default = 1.0)

CBUSH NASTRAN interface Elements

CBUSH Force Output:

$$F(\text{CBUSH}) = K_{\text{dof}_i} * [U(\text{node2})_{\text{dof}_i} - U(\text{node1})_{\text{dof}_i}]$$

Where

- K_{dof_i} = Stiffness defined in PBUSH/CBUSH card for the degree of freedom i.
- $U(\text{node1})_{\text{dof}_i}$ = displacement of the node 1 in the direction of the dof 1, according to **the CBUSH coordinate system**
- $U(\text{node2})_{\text{dof}_i}$ = displacement of the node 2 in the direction of the dof 1, according to **the CBUSH coordinate system**

REMINDER:

- CBUSH force sign depends on how the nodes have been ordered in defining the element.
- CBUSH force direction is defined through the CBUSH coordinate system

Recommendation for CBUSH creation:

- Define the CBUSH by using the option of a coordinate system CID
- CBUSH null length
- Directions of the CBUSH coordinate system must have physical sense. (f.i: bolt tension and shear directions)
- Identical criteria choosing node 1 (structure A) and node 2 (structure B) in all the interface CBUSH elements (easier force post-processing)

MPC's and RBEs: Degrees of Freedom Relations

The explicit MPC's establish relations between the DOF of the nodes by means of a general equation provided by the user.

The RBEs or rigid elements are particular cases of MPC. There are different types of RBE elements: RROD, RBAR, RBE1, RBE2 & RBE3. The most used RBE elements are RBE2 & RBE3.

The main functions of the MPC and RBEs are:

- To establish kinematic relationships between the DOF of several nodes by means of a general equation (explicit MPC)
- To model roughly some part of the structure whose stiffness is much higher than the adjacent components and it is a load path. (RBE2 elements)
- To distribute the load applied on 1 node to several adjacent ones, but without adding any stiffness to the structure (f.i: aerodynamic loads or inertia loads obtained in control points that must be applied on the structure) (RBE3 elements)
- To define the displacements of an isolated node as the average value of other nodes without adding any stiffness to the structure (f.i: representative nodes of the movement of a section) (RBE3 elements)
- To avoid NASTRAN numeric singularities in the execution process due to the lack of stiffness in some DOF

Concept of Degree of Freedom (DOF): singularities in NASTRAN execution

Each node has 6 degrees of freedom (DOF). 3 translation & 3 rotations.

MPC and RBE element establish relationships between the DOF of different nodes. There are independent & dependent DOF in the equations imposed by the MPC and RBE elements.

The dependent DOF must be defined completely and in a unique way by the independent DOF

— NASTRAN singularities in execution process:

- All the DOF of the whole FEM must have stiffness: if a DOF is not defined completely with its associated stiffness of kinematic relationship, a numeric singularity will appear and NASTRAN analysis will not run.

— Limitations and consistency of MPC & RBE DOF nodes relationships:

- A dependent DOF must be defined in a unique way: If 2 different MPC, RBE or SPC are defining the same DOF, an error message will show up and NASTRAN analysis will not run.

Explicit MPC: Definition & typical usage

MPC Multipoint Constraint

Defines a multipoint constraint equation of the form

$$\sum_j A_j u_j = 0$$

where u_j represents degree-of-freedom C_j at grid or scalar point G_j .

Format:

1	2	3	4	5	6	7	8	9	10
MPC	SID	G1	C1	A1	G2	C2	A2		
		G3	C3	A3	-etc.-				

Field	Contents
SID	Set identification number. (Integer > 0)
Gj	Identification number of grid or scalar point. (Integer > 0)
Cj	Component number. (Any one of the Integers 1 through 6 for grid points; blank or zero for scalar points.)
Aj	Coefficient. (Real; Default = 0.0 except A1 must be nonzero.)

Remarks:

1. Multipoint constraint sets must be selected with the Case Control command MPC = SID.
2. The first degree-of-freedom (G1, C1) in the sequence is defined to be the dependent degree-of-freedom. A dependent degree-of-freedom assigned by one MPC entry cannot be assigned dependent by another MPC entry or by a rigid element.

- The DOF U_{G1} (dependent) of the node G1 is defined based on the above equation and the movement of the independent DOF of the other nodes.

$$U_{(G1)} = - \sum f_i * U_{(Gi)}$$

- The DOF are defined in the analysis coordinate system of each node. It is recommended that all the nodes of the MPC should have the same analysis system in order to avoid errors.
- It is strongly recommended to check that the MPC user-defined equations have physical sense.

CAUTION using MPC between non-coincident nodes

- Explicit MPC can be activated / deactivated in the NASTRAN case control section (MPC= ID)

— MPC typical usages:

- To define the movement of a node based on other ones (f.i: to ensure displacement consistency in the junction of 2 componentes modelled with different FEM mesh sizes)
- To define different junction types between 2 structures and to activate the proper one for each NASTRAN analysis

Main RB elements: Definition & typical usage – RBE2

— RBE2 definition:

RBE2 Rigid Body Element, Form 2

Defines a rigid body with independent degrees-of-freedom that are specified at a single grid point and with dependent degrees-of-freedom that are specified at an arbitrary number of grid points.

Format:

1	2	3	4	5	6	7	8	9	10
RBE2	EID	GN	CM	GM1	GM2	GM3	GM4	GM5	
	GM6	GM7	GM8	-etc.-	ALPHA				

Field	Contents
EID	Element identification number. (0 < Integer < 100,000,000)
GN	Identification number of grid point to which all six independent degrees-of-freedom for the element are assigned. (Integer > 0)
CM	Component numbers of the dependent degrees-of-freedom in the global coordinate system at grid points GMi. (Integers 1 through 6 with no embedded blanks.)
GMi	Grid point identification numbers at which dependent degrees-of-freedom are assigned. (Integer > 0)
ALPHA	Thermal expansion coefficient. See Remark 11. (Real ≥ 0.0 or blank)

- The DOF are defined in the analysis coordinate system of each node.
- RBE2 behaves as a rigid solid with an “infinite” stiffness compare with the adjacent structure
- Perfect rigid solid: selected dependent DOF 123456 (CM)

Example: \$2345678. 2345678. 2345678. 2345678. 2345678. 2345678. 2345678.

RBE2 6910000 600000 123456 600630 601240 601250

— RBE2 typical usage:

- To model roughly some part of the structure, whose own detailed stress distribution is not the study objective, but it is a load path and it has a stiffness much higher than the adjacent components (f.i: fittings in a GFEM)

Main RB elements: Definition & typical usage – RBE3

RBE3 Interpolation Constraint Element

Defines the motion at a reference grid point as the weighted average of the motions at a set of other grid points.

Format:

	1	2	3	4	5	6	7	8	9	10
RBE3	EID		REFGRID	REFC	WT1	C1	G1,1	G1,2		
	G1,3	WT2	C2	G2,1	G2,2	-etc.-	WT3	C3		
	G3,1	G3,2	-etc.-	WT4	C4	G4,1	G4,2	-etc.-		
	"UM"	GM1	CM1	GM2	CM2	GM3	CM3			
		GM4	CM4	GM5	CM5	-etc.-				
	"ALPHA"	ALPHA								

Field	Contents
EID	Element identification number. Unique with respect to other rigid elements. (0 < Integer < 100,000,000)
REFGRID	Reference grid point identification number. (Integer > 0)
REFC	Component numbers at the reference grid point. (Any of the integers 1 through 6 with no embedded blanks.)
WTi	Weighting factor for components of motion on the following entry at grid points Gi,j. (Real)
Ci	Component numbers with weighting factor WTi at grid points Gi,j. (Any of the integers 1 through 6 with no embedded blanks.)
Gi,j	Grid points with components Ci that have weighting factor WTi in the averaging equations. (Integer > 0)

Typical RBE3 definition:

Example: \$2345678. 2345678. 2345678. 2345678. 2345678. 2345678. 2345678. 2345678

RBE3 6930000 693000 123456 1.0 123 600630 601240

→ dependent node: DOF 123456 must be defined (in order to avoid NASTRAN singularities)

→ Independent nodes: **only** DOF 123 must be used. (if the DOF 456 are used, NASTRAN will provide results but they will be incorrect)

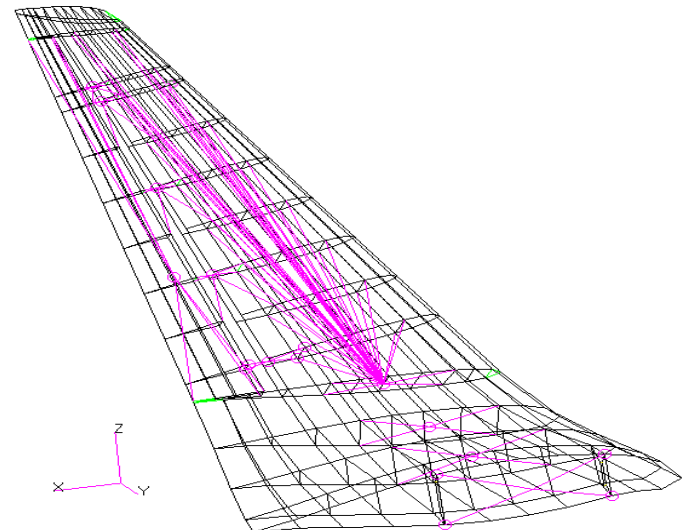
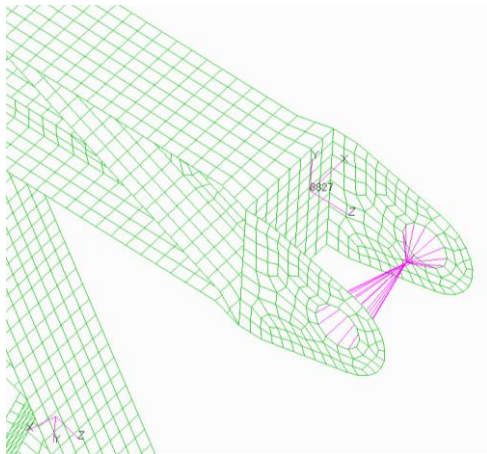
- The DOF 4 or 5 or 6 of the independent nodes **only** must be used in order to avoid singularities, when all the nodes are positioned in a straight line.

- The DOF are defined in the analysis coordinate system of each node.
- RBE3 does not add any stiffness to the structure.
- RBE3 load repartition/average displacement is based on the geometrical positions of the different nodes and on the weighed factors defined in the RBE3 card. It is not based on the adjacent structure stiffness

Main RB elements: Definition & typical usage – RBE3

— RBE3 typical usage:

- To distribute the load applied on 1 node (dependent node) to several adjacent ones (independent nodes), but without adding any stiffness to the structure (f.i: aerodynamic loads or inertia loads obtained in control points that must be applied on the structure)
- To define the displacements of an isolated node (dependent node) as the average value of other nodes (independent nodes) without adding any stiffness to the structure (f.i: representative nodes of the movement of a section)

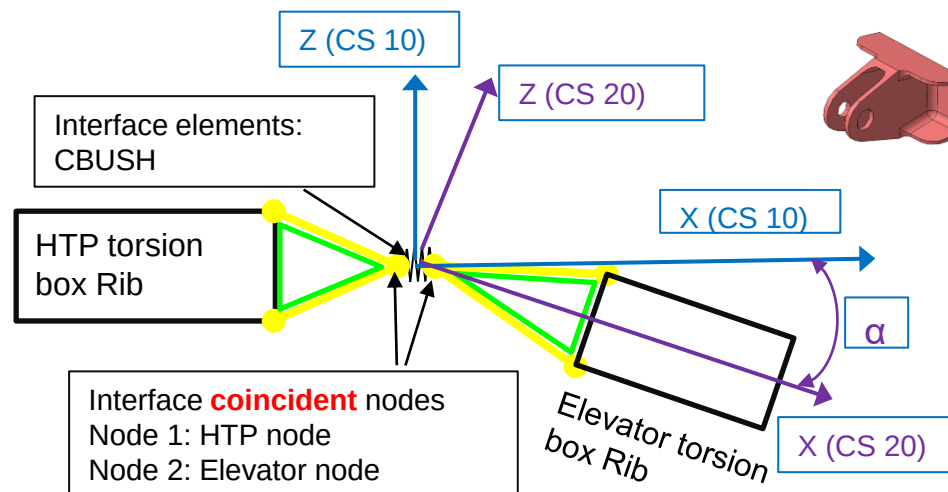


FEM interfaces: Application to Movable Surfaces modelling – Elevator example

The movable surface must be analysed for any position of its rotational angle envelope.

— Elevator FEM example: One Elevator FEM file for each rotational position

- Coordinate system (CS) 10 represents the non-rotated elevator position (Y hinge line direction)
- **CS 20 orientation is defined in NASTRAN with respect to the CS 10**
- **All elevator nodes (only them)** must have the CS 20 as **Reference & Analysis system**.
- The whole Elevator FEM is saved in a separate bdf. It will be called with an INCLUDE sentence
There will be different Elevator FEM files, with only changing the angle α parameter.
- Interface elements: 1 CBUSH in X direction & 1 CBUSH in Z direction, being CBUSH CS 20. One interface fitting will have also 1 CBUSH in Y direction if has been designed for it (CS25.393)
- HTP interface node 1 analysis coordinate system: As CBUSH element is used, nod 1 analysis coordinate system is not relevant. If CELAS are used, its analysis system must be as in node 2
- Interface fittings defined with **CROD** and **CTRIA**
- Pressure loads on the Elevator can be applied either with PLOAD cards or with FORCE/MOMENTS cards. If FORCE/MOMENTS cards types are used, its definition CS must account for the Elevator rotated position (f.i: it must be defined with its orientation respect to CS20)

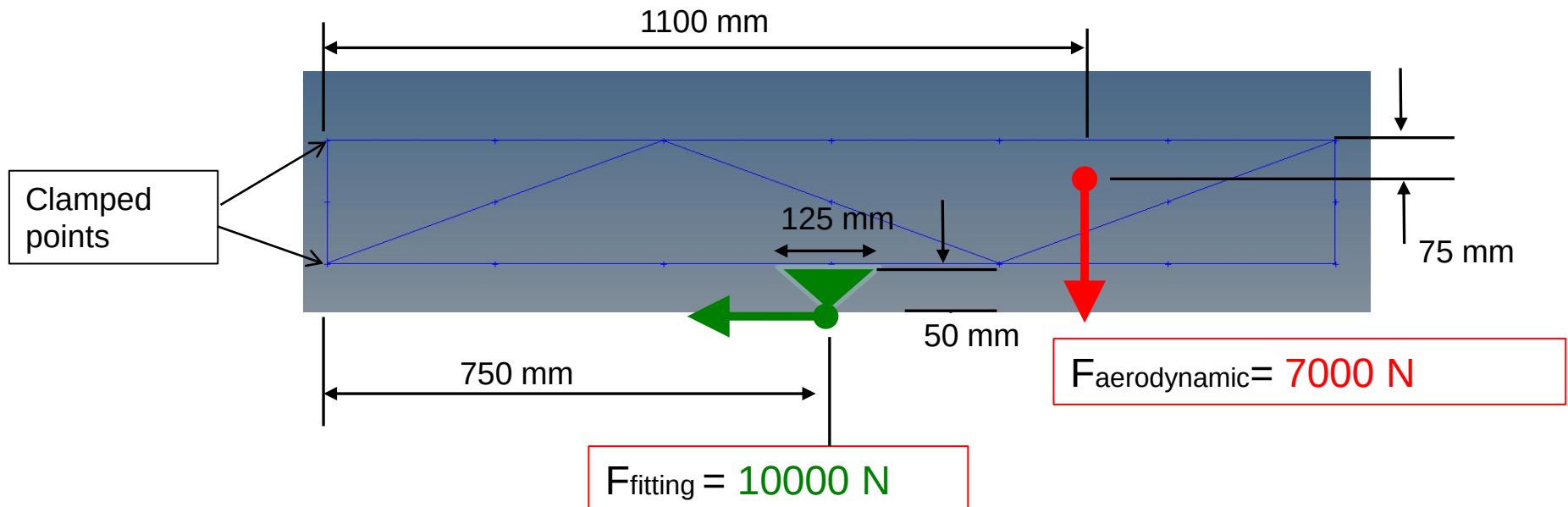


Exercise

Objective

In the structure created for the exercise of Lecture 12:

- To model the stiffness of the clamped points by using CBUSH elements
 $K_x=K_y=K_z=10.000$; $K_{rot_x}=K_{rot_y}=K_{rot_z}=100.000$
- To apply an aerodynamic load resultant by means of RBE3 elements
- To model a rigid local fitting that will be used in order to introduce a local load in the beam structure (the objective of this analysis will be to assess the global impact of this fitting load, and not its local stress analysis)



References

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7. MSC/NASTRAN, "Basic Dynamic Analysis User's Guide"
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